Design and Analysis of a Miller-Compensated Two-Stage Op-Amp

Project Title

Design and Analysis of a Miller-Compensated Two-Stage Op-Amp

Objectives of the Project

- To design a two-stage CMOS operational amplifier in 90nm CMOS technology, consisting of a differential input stage and a common-source gain stage.
- To achieve high open-loop DC gain with relatively good bandwidth and phase margin at low bias currents.
- To characterize the amplifier's differential-mode gain, common-mode gain, phase margin, slew rate, gain-bandwidth, and power dissipation through simulation.
- To validate the schematic-level design and extract performance metrics using Cadence Virtuoso.

Design Specifications

Parameter	**Target Value**
Technology	90nm CMOS (gpdk090)
Supply Voltage (VDD)	1.2 V (0 V to 1.2 V swing)
DC Gain (Av)	52 dB
Phase Margin	78°
Slew Rate	5 V/µs
Power Dissipation	252 µW
Load Capacitance (CL)	2 pF
CMRR	High
Bias Current	20 µA

Design Methodology

- Topology selection – A two-stage architecture is chosen: a PMOS/NMOS differential pair as the first stage followed by a common-source second stage.
- Calculations – Initial sizing of transistors (W/L ratios) is derived using MOSFET square-law equations from the target gain and slew rate specifications, using 90nm process parameters.
- Compensation network design – A Miller capacitor (Cc) is placed between the first and second stage outputs to split the dominant and non-dominant poles.
- Schematic entry and simulation – The circuit is entered and simulated in Cadence Virtuoso to verify DC operating points, open-loop gain/phase and slew rate.
- Iterative tuning – Transistor sizes, bias currents and Cc are iteratively adjusted to meet all specifications simultaneously.
- Corner and parametric analysis – Gain, phase margin, slew rate and power dissipation are verified across process/voltage/temperature corners and bias sweeps to confirm robustness.
- Result extraction – Final DC gain, phase margin, slew rate, and power dissipation are extracted from AC and transient simulation results in Cadence Virtuoso.

EDA Tools / Software to be Used

- Cadence Virtuoso – schematic capture and simulation environment
- ADE-L – DC, AC, and transient simulation
- CMOS technology (gpdk090)

Abstract

An operational amplifier is considered to be the fundamental block in mixed signal and analog design. This project aims to design a two stage CMOS operational amplifier using Miller Compensation technique, mainly to achieve high gain with relatively good bandwidth and phase margin with low bias currents in 90 nm technology. Keeping this as a main aspect, opamp specifications are taken into account, i.e., differential mode gain, common mode gain, phase margin, slew rate, power dissipation and gain bandwidth. This design operates at 1.2V to 0V half voltage supply swing. The simulation process is carried out in Cadence Virtuoso in 90nm technology. The obtained DC gain is 52 dB with a phase margin of 78 degrees and a slew rate of 5 V/µSec with a power dissipation of 252 dµW.

Calculations

Specifications

DC gain = 60 dB

GBW = 30 MHz

Phase margin ≥ 60°

Slew rate = 20 V/µs

ICMR+ = 1.6 V, ICMR− = 0.8 V

CL = 2 pF

Power ≤ 300 µW

Given data

µnCox = 300 µA/V², µpCox = 60 µA/V²

|Vt3|max = 0.51 V, Vt1min = 0.47 V, Vt1max = 0.59 V

I0 = 20 µA, Cc = 600 fF, Rz = 1 kΩ

Transistor sizes (from schematic)

M1, M2 (NM0, NM1): W = 3 µm, L = 500 nm, W/L = 6

M3, M4 (PM2, PM0): W = 7 µm, L = 500 nm, W/L = 14

M5 (NM3): W = 6 µm, L = 500 nm, W/L = 12

Bias transistor (NM4): W = 6 µm, L = 500 nm, W/L = 12

M6 (PM3): W = 20 µm, L = 115 nm, W/L = 173.9

M7 (NM2): W = 30 µm, L = 400 nm, W/L = 75

Calculations

Step 1: Bias currents

I5 = I0 × (W/L)5 / (W/L)bias = 20 µ × 12 / 12 = 20 µA

ID1 = ID2 = I5/2 = 10 µA

I3 = I4 = 10 µA

I6 = I4 × (W/L)6 / (W/L)4 = 10 µ × 173.9 / 14 = 124.2 µA

I7 = I5 × (W/L)7 / (W/L)5 = 20 µ × 75 / 12 = 125 µA

Step 2: Slew rate and Cc

SR = I5 / Cc = 20 µA / 600 fF = 33.3 V/µs

Cc ≥ 0.22 CL = 0.22 × 2 pF = 440 fF

Step 3: Input pair M1, M2

gm1 = 2π × GBW × Cc = 2π × 30 MHz × 600 fF = 113.1 µS

(W/L)1,2 = gm1² / (µnCox × I5) = (113.1 µ)² / (300 µ × 20 µ) = 2.13

For (W/L)1,2 = 6 (used in schematic):

gm1 = sqrt(2 × µnCox × (W/L)1 × ID1) = sqrt(2 × 300 µ × 6 × 10 µ) = 189.7 µS

GBW = gm1 / (2π Cc) = 189.7 µ / (2π × 600 f) = 50.3 MHz

Step 4: M3, M4 from ICMR+

ICMR+ = VDD − sqrt(2 I3 / β3) − |Vt3|max + Vt1min

VDD − ICMR+ − |Vt3|max + Vt1min = 1.8 − 1.6 − 0.51 + 0.47 = 0.16 V

(W/L)3 = 2 I3 / [μpCox × (0.16)²] = 20 μ / (60 μ × 0.0256) = 13.02, taken as 14

sqrt(2 I3 / β3) = sqrt(2 × 10 μ / (60 μ × 14)) = 0.1543 V

ICMR+ = 1.8 − 0.1543 − 0.51 + 0.47 = 1.606 V

Step 5: M5 from ICMR−

Vov1 = sqrt(2 ID1 / (μnCox (W/L)1)) = sqrt(2 × 10 μ / (300 μ × 6)) = 0.1054 V

Vdsat5 ≥ ICMR− − Vov1 − Vt1max = 0.8 − 0.1054 − 0.59 = 0.1046 V

Vdsat5 = sqrt(2 I5 / (μnCox (W/L)5)) = sqrt(2 × 20 μ / (300 μ × 12)) = 0.1054 V

ICMR− = Vov1 + Vt1max + Vdsat5 = 0.1054 + 0.59 + 0.1054 = 0.801 V

Step 6: M6

gm4 = sqrt(2 × μpCox × (W/L)4 × I4) = sqrt(2 × 60 μ × 14 × 10 μ) = 129.6 μS

gm6 = sqrt(2 × μpCox × (W/L)6 × I6) = sqrt(2 × 60 μ × 173.9 × 124.2 μ) = 1610 μS

(W/L)6 = (gm6 / gm4) × (W/L)4 = (1610 / 129.6) × 14 = 173.9

Step 7: M7

(W/L)7 = (I7 / I5) × (W/L)5 = (125 / 20) × 12 = 75

Step 8: Systematic offset

(W/L)6 / (W/L)4 = 2 (W/L)7 / (W/L)5

LHS = 173.9 / 14 = 12.42

RHS = 2 × 75 / 12 = 12.5

Step 9: Poles, zero and phase margin

p2 = gm6 / CL = 1610 μ / 2 p = 8.05 × 10^8 rad/s, f(p2) = 128 MHz

RHP zero without Rz: z = gm6 / Cc = 1610 μ / 600 f = 2.68 × 10^9 rad/s, f(z) = 427 MHz

PM (without Rz) = 90° − $\tan^{-1}$(GBW/z) − $\tan^{-1}$(GBW/p2) = 90° − 6.7° − 21.4° = 61.8°

1/gm6 = 621 Ω

With Rz = 1 kΩ: z = 1 / [Cc (Rz − 1/gm6)] = 1 / (600 f × 379) = 4.40 × 10^9 rad/s, f(z) = 700 MHz (LHP)

PM (with Rz) = 90° + $\tan^{-1}$(GBW/z) − $\tan^{-1}$(GBW/p2) = 90° + 4.1° − 21.4° = 72.7°

Step 10: Power

Itotal = I0 + I5 + I7 = 20 + 20 + 125 = 165 μA

P = VDD × Itotal = 1.8 × 165 µA = 297 µW

Step 11: Output swing

Vdsat6 = sqrt(2 I7 / (µpCox (W/L)6)) = sqrt(2 × 125 µ / (60 µ × 173.9)) = 0.155 V

Vdsat7 = sqrt(2 I7 / (µnCox (W/L)7)) = sqrt(2 × 125 µ / (300 µ × 75)) = 0.105 V

Vout(max) = VDD − Vdsat6 = 1.645 V

Vout(min) = Vdsat7 = 0.105 V

Summary

I5 = 20 µA, ID1 = 10 µA, I6 = 124.2 µA, I7 = 125 µA

SR = I5 / Cc = 33.3 V/µs

gm1 = sqrt(2 µnCox (W/L)1 ID1) = 189.7 µS

GBW = gm1 / (2π Cc) = 50.3 MHz

ICMR+ = VDD − sqrt(2 I3/β3) − |Vt3|max + Vt1min = 1.606 V

ICMR− = Vov1 + Vt1max + Vdsat5 = 0.801 V

gm6 = sqrt(2 µpCox (W/L)6 I6) = 1610 µS

A0 = gm1 (ro2 ∥ ro4) × gm6 (ro6 ∥ ro7) = 52 dB (simulated)

p1 = GBW / A0 = 83 kHz (simulated)

p2 = gm6 / CL = 128 MHz

z = 1 / [Cc (Rz − 1/gm6)] = 700 MHz

PM = 90° + $\tan^{-1}$(GBW/z) − $\tan^{-1}$(GBW/p2) = 72.7°

P = VDD × (I0 + I5 + I7) = 297 µW

Vout(min) = 0.105 V, Vout(max) = 1.645 V